Molecular Dynamics Simulation of a Lennard-Jones Liquid

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Description

The program [1] in this repository simulates a Lennard-Jones Liquid with variables: $N = 100$ particles in a 2D square box with side, $L = 11$ ($\Rightarrow \rho = 0.83$), with velocities distributed as per $T = 1$

Dependencies

I use the **Raylib** Graphics Library v5.5-1 for the interface and **gnuplot** v6.0 for the graphs.

How to run

Look for the file `ljliquid.c`, get Raylib 5.0 or higher, then compile using the command

```
$> gcc ljliquid.c -lm -lraylib -o ljliquid
```

This should produce a binary that can be run in the terminal like

```
$> ./ljliquid
```

Plots

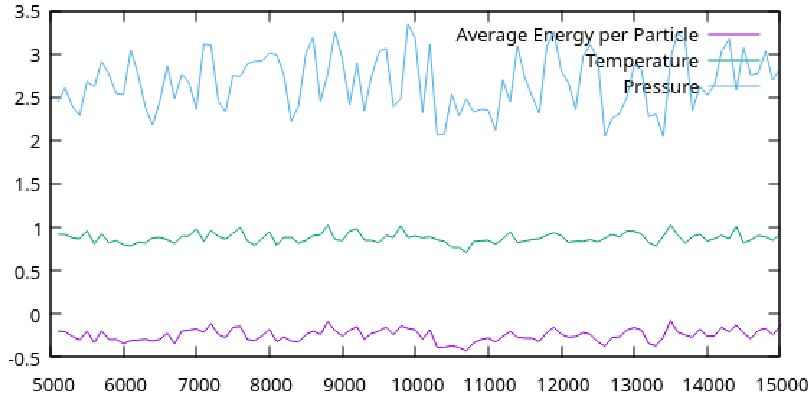

Figure 1: Graph of energy, temperature and pressure

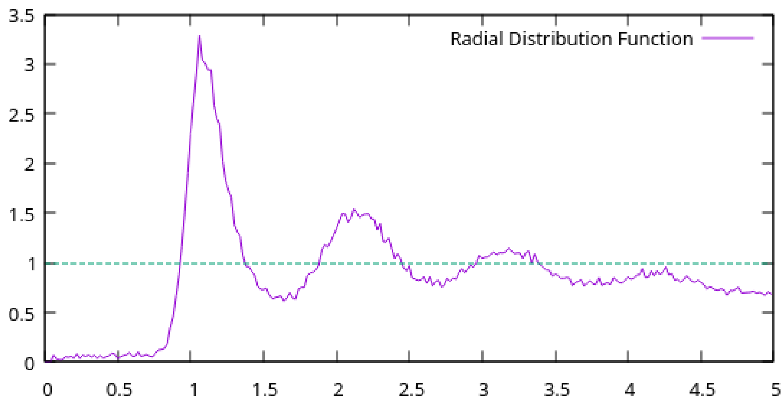

Figure 2: Radial Distribution Function (rdf) sampled during the production run

Principles used

Using the velocity Verlet algorithm, we get a order Δt^4 accuracy in position, Δt^3 in velocity and Δt^2 in Energy[2]:

- Calculate $\mathbf{x}(t+\Delta t) = \mathbf{x}(t) + \mathbf{v}(t)\,\Delta t + \frac{1}{2}\,\mathbf{a}(t)\,\Delta t^2$.
- Derive $\mathbf{a}(t+\Delta t)$ from the interaction potential using $\mathbf{x}(t+\Delta t)$.
- Calculate $\mathbf{v}(t+\Delta t) = \mathbf{v}(t) + \frac{1}{2}\left(\mathbf{a}(t) + \mathbf{a}(t+\Delta t)\right)\Delta t$.

Observations

The rdf of the system looks like that of liquids. So, we can assume under the given conditions (T = 1, ρ = 8.3), any particles with the Lennard-Jones potential will be in the liquid state.

References

[1] Repository containing ljliquid.c on Github

[2] Velocity Verlet on Wikipedia